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Reconstructing North Atlantic Ocean Heat Content Using Convolutional Neural Networks

Master Thesis

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Abstract

Slowly varying ocean heat content is one of the most important variables when describing the Earth system's memory on interannual to decadal time scales. Numerical climate models simulate the evolution of oceanic heat content after incorporating a set of observations via model assimilation. However, incomplete observations, particularly in the subsurface ocean constrain the accuracy of model assimilation and lead to large uncertainties. Kadow et al (2020) have proven that artificial intelligence can successfully be utilized to reconstruct missing climate information in the form of surface temperatures. In the following, I investigate the possibility to train their neural network to reconstruct missing ocean heat content data.

To perform the reconstruction, I use a three-dimensional convolutional neural network originally designed for image inpainting technology. Combining artificial intelligence with earth system modeling, the network is trained and tested with a 16 member ensemble model assimilation from the Max-Planck Institute Earth System Model (MPI-ESM). These model runs are masked mirroring real life observations and can be adapted in their missing value density to the reconstruction time period in question. The monthly training and testing datasets, which are fed individually to neural network, run from January 1958 till December 2020 and thus incorporate an additional time dimension into the training process.

1 Introduction

1.1 Motivation and Problem

1.1.1 Why is OHC important to the climate system?

Predicting changes of Earth system variables on interannual to decadal scales has been the focus of large amounts of current climate research. From forecasts of extreme weather events like droughts or heat waves to predictions of precipitation and temperature on regional scales, decadal predictions yield a number of important results (Marotzke et al., 2016; Keenlyside and Ba, 2010). However, land surface and atmospheric components of the Earth system often change so quickly that they are difficult to rely on when predicting on larger timescales from months to decades. Due to their low frequency variability and long memory in comparison to the atmosphere, oceans play a special role in climate modeling on interannual to decadal time scales. In particular ocean heat content indicates long term changes in heat fluxes between the Earth system's oceanic and atmospheric components (Cheng et al., 2017). In addition to that, it also shows a consistent and slowly varying response to changes in external boundary conditions of the climate system. Therefore, representing changes in ocean heat content and its response to external forcings accurately is one of the major challenges for decadal climate predictions.

1.1.2 What happens with North Atlantic OHC – especially SPG – with progressing climate change?

1.1.3 Current Literature on North Atlantic OHC estimates

Ocean heat content is not only important for climate predictions due to its low frequency variability. Additionally, it is instrumental to the Earth's energy budget and its response to anthropogenic greenhouse gas emissions. Greenhouse gases reflect longwave radiation back towards the Earth's surface, creating an imbalance in the Earth's energy budget in the form of a surplus of incoming energy compared to the outgoing longwave radiation. More than 90% of this energy imbalance is retained in the global oceans with over 30% being stored in the North Atlantic (Häkkinen et al., 2015). This does not only lead to thermal expansion and thus a direct rise in sea level but also affects the formation of sea ice and the global circulation systems. In comparison to surface land and sea temperatures, changes in ocean heat content also show lower background noise in comparison to their signal, making them a better indicator for climatic changes (Cheng et al., 2018). While the overall global heat content has risen significantly over the past 50 years due to anthropogenic climate change, regional differences can be observed. The subpolar North Atlantic for example, or more specifically the Subpolar Gyre (SPG), has seen an overall cooling trend (SOURCE). SOURCE attribute these changes to regional natural variability closely connected to the North Atlantic Oscillation (NAO), which impacts the regional climate in Europe as well as northern North America (SOURCE). The North Atlantic

and especially its sea surface temperature (SST) can be predicted with relatively high skill on interannual to decadal timescales (Buckley et al., 2019). However, especially when examining subsurface temperatures in the North Atlantic and OHC, several sources of uncertainties remain. Better estimating these uncertainties could thus give insights into the different regional OHC trends in the North Atlantic and their impact on European climate and weather.

Additionally, most Earth system models (ESMs) require a current state close to observations as an initialization. Thus, accurate climate predictions on interannual to decadal time scales depend both on the initial state of ocean heat content and its response to external forcing (Yeager et al., 2012).

1.1.4 Uncertainties on OHC esimation

In general, ocean heat content estimates are dependent on several factors. A first and very direct source of uncertainties is the bias measurement instruments bring into the calculation of ocean heat content. Over large parts of the 20th century, mechanical bathythermographs (MBTs) and expendable bathythermographs (XBTs) provided considerable amounts of ocean temperature profile data. MBTs accounted for 68% of profiles from 1941–1966 before being replaced by XBTs, which then accounted for 38% of profile data from 1970–2001 (Gouretski and Cheng, 2020). Both measurement methods have been found to incorporate a time variant warm bias, which significantly impacted ocean heat content timeseries. Gouretski and Cheng (2020) show that time and spatially dependent corrections can help significantly reduce the MBT bias of 0.2°C. Nonetheless, even after the correction a bias of up to 0.1°C remains, particularly in the southern hemisphere. Similarly, Cheng et al. (2014) have designed a correction method for the XBT bias, which limits the bias up to ±0.04°C. In addition to XBT and MBT biases, Boyer et al. (2016) have identified mapping methods and baseline climatologies as additional sources of uncertainties. Especially since scientific consensus on the choice of XBT bias correction method has been reached by now (Garcia-Soto et al., 2021), Boyer et al. (2016) have identified the choice of mapping method as a larger source of uncertainty for the calculation of ocean heat content.

1.1.5 Mapping methods in current literature

Mapping methods refer to the way of reconstructing a full set of values of a variable from a limited amount of observations. As ocean heat content cannot be observed directly, it is usually calculated by integrating over temperature profiles from the ocean surface down to a depth of up to 2.000 m. However, subsurface temperature observations often lack adequate coverage and thus cannot be used without further processing to initialize climate models. In order to create a complete set of values close to the observed state, a range of statistical mapping methods can be applied. On the one hand, data assimilation runs aim to bring together Earth system model processes with an incomplete observed state. The complete set of values hereby produced can then be used for model initialization. On the other hand, statistical methods interpolate between missing data points on a purely mathematical basis. Both techniques have shown high correlation to each other in the reconstruction of oceanic heat content Brune et al.

(2018). However, data assimilation requires high computational costs and statistical infilling methods can have trouble reproducing physical processes in undersampled regions. Additionally, data assimilation runs show lower correlation to interpolation methods in the second half of the twentieth century compared to assimilations run over the last 20 years (Brune et al., 2018).

1.1.6 Dependence of mapping methods on data coverage

1.1.7 The argo/preargo uncertainty problem: Definition of uncertainty and observation density

At the start of the 2000s the amount of subsurface temperature observations in global oceans started to rise rapidly due to the introduction of the Argo project. The Argo project is an internationl cooperation, which placed 4.000 swimming buoys in the global oceans (Roemmich et al., 2009). These buoys drift with ocean currents and continually measure temperatures up to a depth of 2.000 m. The timeseries of different infilling methods of global upper ocean heat content show much higher correlation with each other after the introduction of Argo floats, as can be seen in figure 1.1. As there is no reference point for the ground truth available at points without direct observation, the only estimate of the missing data comes in form of uncertainty ranges of different infilling methods. Therefore, the uncertainty range of ocean heat content estimates at locations with undersampled temperature measurements is much higher before the introduction of Argo floats (Levitus et al., 2012).

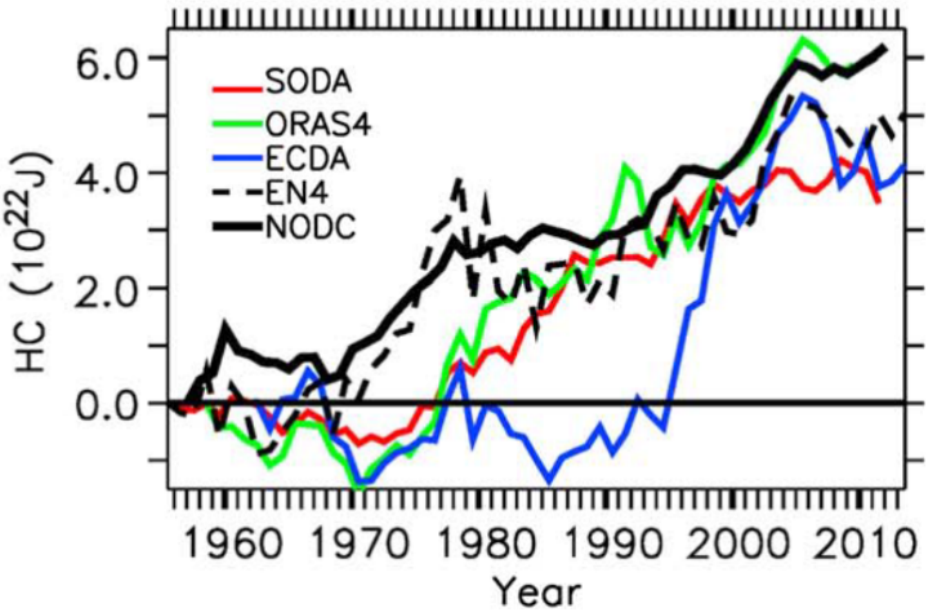

Figure 1.1: Estimates of upper 2.000 m North Atlantic ocean heat content using different data assimilation and interpolation methods (Häkkinen et al., 2015).

1.2 My Approach to a better solution to the problem

1.2.1 What alternative ways to reconstruct OHC? Directer? Faster?

1.2.2 Aim 1: Better defining NA OHC uncertainty with ML reconstructions

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1.3 Summary and Thesis Structure

2 Data and Methods

2.1 Data

In order to gain a new estimate of North Atlantic ocean heat content, this work will use two different sources of training and testing data for the utilized U-net. On the one hand, data assimilation runs from the Max Planck Institute Earth System Model (MPI-ESM, low resolution, Mauritsen et al. (2019)) are deployed to train the neural network to reconstruct temperature maps. On the other hand, observational data from the fourth version of the Met Office Hadley Center 'EN' series (EN4, Good et al. (2013)) subsurface ocean temperature profiles are applied as basis for the final reconstruction. After the neural network reconstructions, the temperature profiles are reverted back to total values and converted into a two dimensional OHC map via integration:

$$H = c_p \int_{z_0}^{z_1} \rho_z \cdot T_z dz, \tag{2.1}$$

In this case H denotes the ocean heat content, c_p the specific heat capacity of sea water, ρ_z the density of the water at depth z and T_z the temperature at this depth. This OHC field is then reduced to one value by summing over all relevant gridpoints for the calculation of an OHC timeseries.

2.1.1 Data Assimilation

In general, Kalman Filter (KF) assimilations assume the uncertainties of model and observations to be Gaussian and then assimilates the observations sequentially. Both, model and observations, are described as probability density functions (PDFs) and the Kalman filter alternates between forecasting and analyzing the PDF's covariance and mean Carrassi et al. (2018). It is often supplemented by a backward calculation of the smoothing mean and covariance to refine the previous estimates in what is called the Kalman smoother (KS). By introducing an ensemble of model trajectories instead of one single representation of the model dynamics, the KF can be adjusted and extended in what is called an ensemble Kalman filter (EnKF). This allows for the covariance matrices to be approximated by an ensemble of model states. The covariance matrix and ensemble mean are then sequentially forecast and updated. EnKFs provide computationally efficient and very accurate results, especially in high dimensional systems, such as the climate and Earth system (Carrassi et al., 2018).

For this work, an ensemble coupled MPI-ESM assimilation (Brune et al., 2021) is used to train the U-net to reconstruct masked temperature profiles. An oceanic localized EnKF, implemented with the Parallel Data Assimilation Framework (Nerger and Hiller, 2013), assimilates monthly EN4 temperature and salinity profiles. Simultaneously, atmospheric nudging of

ERA40/ERAInterim/ERA5 reanalyses (Uppala et al., 2005; Dee et al., 2011; Hersbach et al., 2020) takes place. All monthly datasets range from January 1958 until October 2020 and present temperature profiles up to a depth of 2.000 m in 40 depth steps.
These assimilation runs are based on the work of Brune et al. (2015) and use a version of the singular evolutive interpolated Kalman filter (SEIK, Pham et al. (1998)) in their EnKF. This assimilation was developed as a model inherent initialization for the MPI-ESM and showed improvements especially in the assimilation of sea surface temperatures and ocean heat content (Brune et al., 2015). Therefore, it provides a good estimate of the relevant variables and represents a fitting choice as training data for the U-net.

2.1.2 Observational Data

The EN4 project describes a collection of global ocean temperature and salinity profile datasets spanning from 1900 until present day. It contains observational data from Argo floats, Arctic Synoptic Basinide Oceanography (ASBO), the Global Temperature and Salinity Profile Program (GTSPP) and the World Ocean Database (WOD13) (Good et al., 2013). As these datasets partly overlap, input data is preprocessed and duplicates sorted out before the complete EN4 dataset is created. The same procedure is applied to profiles that are not necessarily identified to be the same but that are very close to each other in time and space (Good et al., 2013). Additionally, various quality control checks are performed in order to identify data profiles with faulty location or measurement data, such as missing depth steps or the location assignment to a point on land rather than in the ocean. All remaining datapoints are then collected in NetCDF files and are publicly available under `https://www.metoffice.gov.uk/hadobs/` (Good et al., 2013). Furthermore, the EN4 project also provides an interpolated version of the observed data points in the form of a reanalysis. These are created by combining a persistence based forecast from the previous month's reanalysis with the observed temperature profiles Good et al. (2013). As the persistence relaxes towards the baseline climatology when not corrected by observations, the EN4 reanalysis tends to posses a lot of uncertainty in times with low observational density. The applied baseline climatology is taken from the years 1970 – 2000. Thus, the overall cooling trend in the NA during that time (**??**) leads to a cool bias especially in the EN4 OHC estimate especially in the remaining pre-Argo era Good et al. (2013). Despite these shortcomings, the EN4 will be taken as a reference point in the evaluation of the network reconstructions as they are highly consistent with the observational data, especially in the well observed Argo era.

2.2 Methods

2.2.1 The Convolutional U-net

The U-net adopted in this work is based on the one used by Kadow et al. (2020) to reconstruct near-surface air temperature data. However, a few modifications have been made for it to fit the task of reconstructing ocean heat content data instead. In general, the U-net is set up to incorporate four encoding and four max pooling layers. These are accompanied by the same amount of decoding and upsampling layers on the decoding path as depicted in figure **??**. Each encoding and decoding layer consists of a partial convolution of the input image. Partial

convolutions use a masked and re-normalized convolution operation, which is then followed by a mask-update step (Liu et al., 2018). In mathematical terms, this means that the partial convolution at each location is defined as

$$x' = \begin{cases} W^T(X \cdot M)\frac{\sum(1)}{\sum(M)} + b, & \text{if } \sum(M) > 0 \\ 0, & \text{otherwise} \end{cases}, \tag{2.2}$$

where W denotes the convolutional filter weights, b its corresponding bias, X the feature values for the current convolution, M the corresponding mask and 1 an array with the same size as M filled with ones Liu et al. (2018). This mask is then updated for each element according to Liu et al. (2018) as

$$m' = \begin{cases} 1, & \text{if} \sum(M) > 0 \\ 0, & \text{otherwise} \end{cases}. \tag{2.3}$$

So, with enough partial convolutions, the mask is eventually updated to incorporate only ones and the output image does not possess any empty pixels anymore. Decoding layers then additionally use the *PyTorch* interpolation function to upsample the image successively back to its original scale.

This whole process is then passed to the forward function of the neural network and conducted for all encoding and decoding layers using ReLU activation functions. Afterwards, the loss is calculated by separating the overall loss function into five parts, each for a different feature of the network output. On the one hand, the loss functions for the infilled holes and originally unmasked data points are weighted individually. On the other hand, their features are also added separately to form the total loss of the output image. Furthermore, a value for the total variation loss is added to account for the variation between neighboring image cells. The individual components are always calculated as mean absolute error (MAE) between the network output and the ground truth of the training data. The weights of the individual loss components are determined manually by tryouts without any specific hyperparameter training. Finally, the network weights are updated using *PyTorch's* backpropagation and optimization functions according to the overall loss function.

2.2.2 Neural Network Setup and Parameters

The U-net conceptually described in section 2.2.1 can be setup in a number of different ways depending on the exact purpose of the training and the input data. In general, the network is trained to reconstruct each slice of the temperature profile separately and then reassembled after the reconstruction. As the aim is to calculate the upper 700 m ocean heat content, the first 20 depth steps of the assimilation training data are selected, which represent temperature fields up to a depth of 700 m. 20 different networks then train to reconstruct the individual depth layers.

The hyperparameters of the training, such as network depth or loss parameters can be adjusted

previous to each training instance. The described setup with four encoding and four pooling layers is taken as a default setting, because until now, a further deepening of the neural network slows down calculations without providing significantly better results. Furthermore, the time period as well as the spatial extension in the form of longitudes and latitudes of the input data can be individually determined for each training process. In regard to the time period of the training data, two different arguments can be adjusted. On the one hand, the origin of the data itself can be adjusted, with default options from he Argo era, pre-Argo era or the entire assimilations time period. On the other hand, the time of origin for the observational patterns, with which the assimilation data is masked during the training process can be changed. Thus, it is possible for the network to learn Argo-era assimilation patterns while also learning to adjust to pre-Argo era observational density.
The remaining parameters, such as learning rate, batch size and image size, are adopted from the original network set up by Kadow et al. (2020) and can be viewed with the remainder of the code under `https://github.com/slenta/Master-Arbeit`.

2.2.3 Data Preprocessing

The U-net described in section 2.2.1 takes three input images in the form of a ground truth assimilation, a binary mask and a masked out assimilation image formatted as an hdf5 file. In order feed the NetCDF4 files containing the assimilation data into the network, they have to be preprocessed to fit the given format. At first, the assimilation's temperature profiles are converted to anomalies using the assimilation's average monthly climatology. For this calculation only the time period from 2004 – 2020 is considered, as this work assumes that the assimilation possesses significantly less uncertainty during the well observed Argo era. Then, EN4 observations from a selected input time are converted to binary masks and multiplied with the data assimilation model data. All three, the masks, the masked data as well as the data assimilations are then stored in an hdf5 file. From there, they can be loaded, split up into batches and fed into the network using a class with the *PyTorch* dataset module.
After the training process, when reconstructing a dataset with monthly observations for a certain timespan, the U-net returns outputs with fully reconstructed monthly temperature profiles.

2.3 Reconstructing North Atlantic Ocean Heat Content

The aim of this work is to develop an improved uncertainty estimate on North Atlantic ocean heat content, specifically in the time before the availability of Argo data. To do this, I propose to divide the available 16 member ensemble of monthly data assimilation runs into two distinct eras. The first part will be called the pre-Argo era, which starts with the beginning of the assimilation runs in 1958 and ends once the Argo project has reached close to global coverage of subsurface temperature observations in 2004. The timespan afterwards, once the Argo floats have reached near global observation coverage, will be called the Argo era and ends with the end of the assimilation runs in October 2020.

To assess, if convolutional U-nets can be considered a valid alternative to EnKF data assimi-

lation, the neural network will be trained on data assimilations during the well observed Argo era (2004 – 2020) in order to learn on assimilations, which posses lower uncertainty and have integrated more observational data. After the network has been fully trained, visible at the progression of its training curve, it can be given masked temperature profiles for reconstruction. At first, the network will reconstruct the Argo era masked assimilations from one of the held back ensemble members. Each month of assimilation data will be masked with the corresponding observational pattern from that time. Given the high observational density during the Argo era, this reconstruction can be considered the first step for the neural network towards the reproduction of the assimilation OHC estimate. This approach can then be extended to the pre-Argo era, if the neural network is able to reproduce the assimilation ensemble within its uncertainty range during the Argo era. Analyzing the reconstructed OHC timeseries as well as maps from individual timesteps, the machine learning reconstructions can also be used to discuss inconsistencies and uncertainties within the assimilation. Differences between both OHC estimates can then be taken as indicator for uncertainties that are not mirrored within the assimilation's ensemble spread. Finally, I will examine, whether the neural network is able to directly reconstruct observations in the pre-Argo and Argo era as well as more recent observations it has not trained on within the assimilations uncertainty range. The results will then be evaluated in regard to the neural network's potential to reproduce or replace EnKF assimilations. Additionally, its implications on a new uncertainty estimate on the EnKF assimilations OHC outside of its ensemble spread will be considered.

3 Results

At first, the U-net is trained with assimilations as well as masks only from the Argo era. Thus, it is set up as well as possible to reproduce argo era assimilations. The training process is then conducted until the loss parameters do not show any significant decrease anymore. Manual selection has shown that the most important loss parameters for the reconstruction of masked temperature profiles are those monitoring the direct misfit of each gridpoint and the total variation loss between neighboring gridpoints. Thus, their multiplicators have been upgraded in this version of the convolutional U-net and their development will be shown to indicate the state of training of the neural network.

Figure **??** shows the development of the hole and valid loss functions of the network training on reconstructing the first depth layer, the sea surface temperatures (SSTs). After around 1000000 training iterations, the network's loss function is not reduced significantly anymore with additional training iterations. It has reached a stable plateau. Thus, the first part of the training process is stopped and followed by 200000 iterations of finetuning. In the process of finetuning a lower learning rate of $5 \cdot 10^{-5}$ ensures that the network moves as close as possible towards the local minimum it would be circling around with a higher learning rate. Figure **??** depicts the development of the validation loss. Validation is conducted by giving the network several timesteps of one of the held back assimilation ensemble members (r2i1p1f1) to reconstruct. The resulting validation error is then calculated parallel to the training error. The progression of the validation loss shows that the network has not adjusted too much to the training data up to 1.000.000 training iterations. Thus, overfitting of the network is prevented.

After the training process is complete, the Argo era assimilation data of another held back ensemble member (r15i1p1f1) is masked with the corresponding observational patterns. These masks are then given to the trained neural network for reconstruction. The reconstructed temperature profiles are then converted to a timeseries of NA OHC. Figure (3.1) shows this timeseries of Argo era NA OHC for both the EnKF assimilations (red) and the neural network reconstructions of masked assimilations (blue). The uncertainty range around both timeseries represents the corresponding ensemble spread. In the case of the neural network, this ensemble is produced by reconstructing the masked assimilations at different training iterations. The network state of each member is extracted 1000 iterations apart from the next. Thus, the overall state of the network is not change, but the uncertainty stemming from the exact moment of ending the training can be estimated.

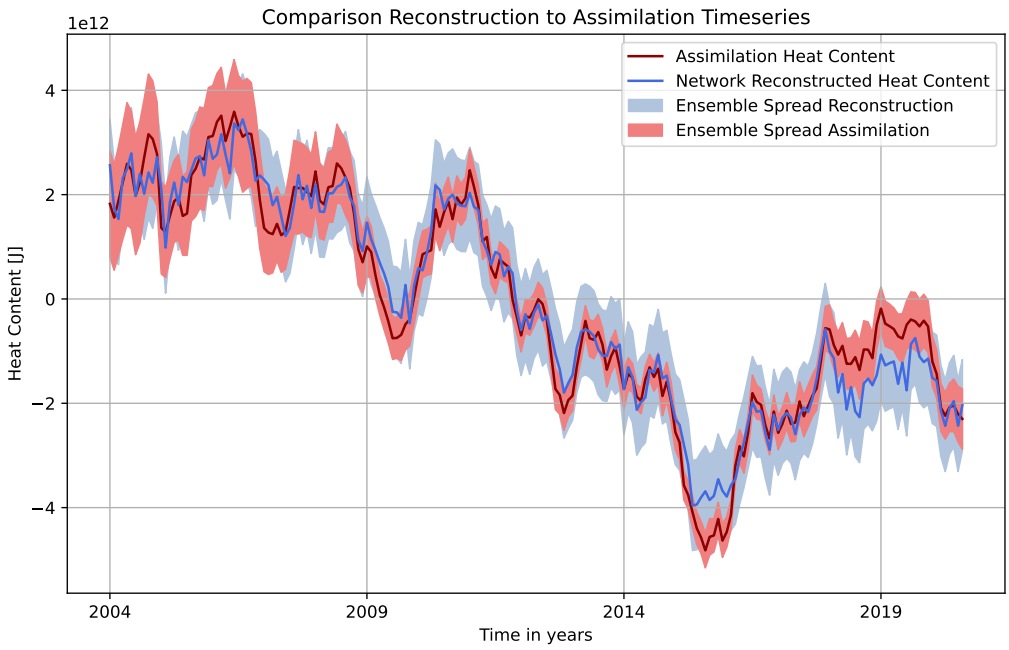

Figure 3.1: Estimates of upper 700 m North Atlantic OHC of one EnKF assimilation's ensemble member (red) and the neural network reconstruction (blue). The shaded areas represent the uncertainty caused by the corresponding ensemble spread.

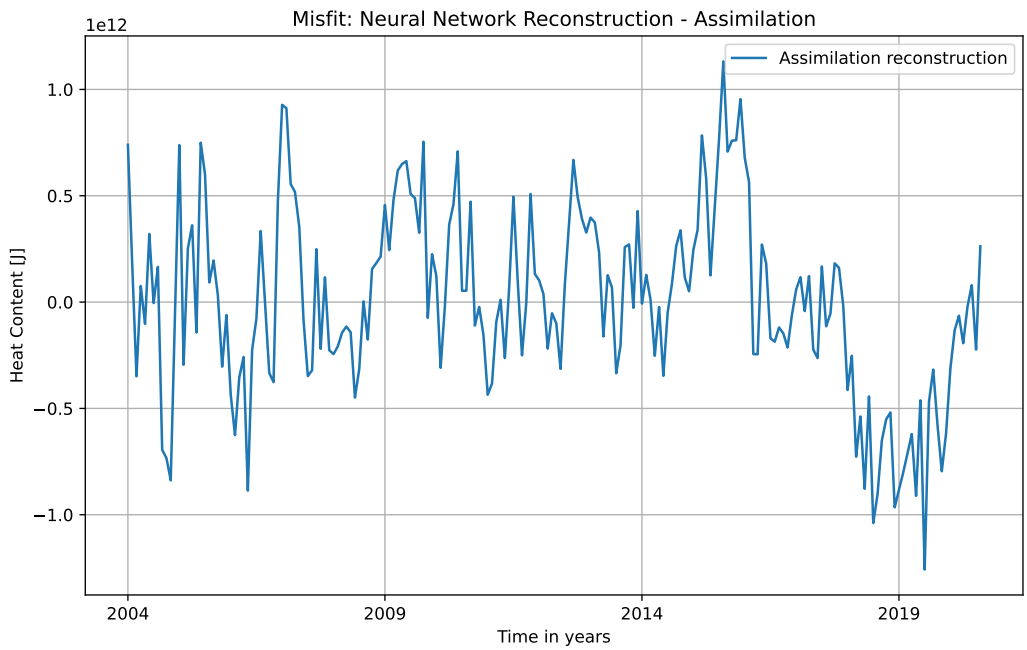

Figure 3.2: Misfit between EnKF assimilation NA upper 700m OHC and the neural network's reconstruction.

Within the Argo era, the neural network is able to reconstruct assimilations masked with observational patterns within the assimilation's uncertainty range for monthly values. Not only

do both uncertainty ranges overlap, the fit is clearly better than just inside of one standard deviation. The neural network's reconstruction even manages to capture monthly trends in the assimilation very well. Analysing this timeseries further, the misfit between both estimates shows that around 2019 the reconstruction shows a slight cool bias compared to the assimilation (3.2). But, as the remaining timeseries does not show any clear bias but rather just errors distributed around zero, it can be assumed to be insignificant and not a sign of a larger uncertainty within the network reconstructions. Figure 3.3 supports that assumption. It depicts NA OHC as calculated by the EnKF assimilation and the neural network for the relevant timespan from August 2018 until April 2019. The images show that no large scale bias exists during that time but that it is rather caused by several smaller scale differences in different regions. Figure 3.3 also shows that not only the network's overall OHC agrees with the EnKF assimilation, but also all major spatial OHC patterns are represented well by the neural network. The cooling of the SPG is well represented as well as the warming of the subtropical gyre (STG) and even such small scale features as the Mediterranean sea.

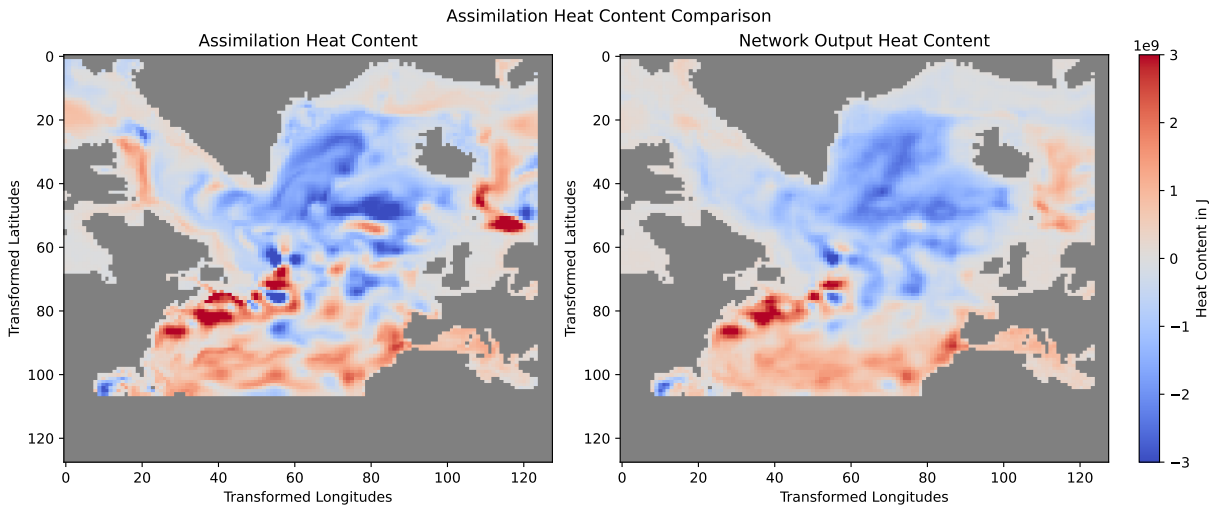

Figure 3.3: Map of EnKF assimilation (left) and neural network reconstruction of NA upper 700m anomaly OHC estimate.

To further analyse the neural network's capability to capture the assimilation's spatial patterns, the pattern correlation between both OHC estimates can be taken. The pattern correlation is calculated by flattening both OHC images to a one-dimensional array and taking the anomaly correlation coefficient of both arrays. Thus, the similarities between spatial patterns for both OHC images can be estimated. Figure 3.4 depicts this pattern correlation during the Argo era. During the first years of the Argo era both OHC estimates show a slightly lower pattern correlation while with rising observations during the Argo era the pattern correlation reaches a constant value of over 0.8. The first 5 years also show higher month-to-month variability in pattern correlation. This can also be attributed to the higher inconsistencies in data coverage during the first years of the Argo project.

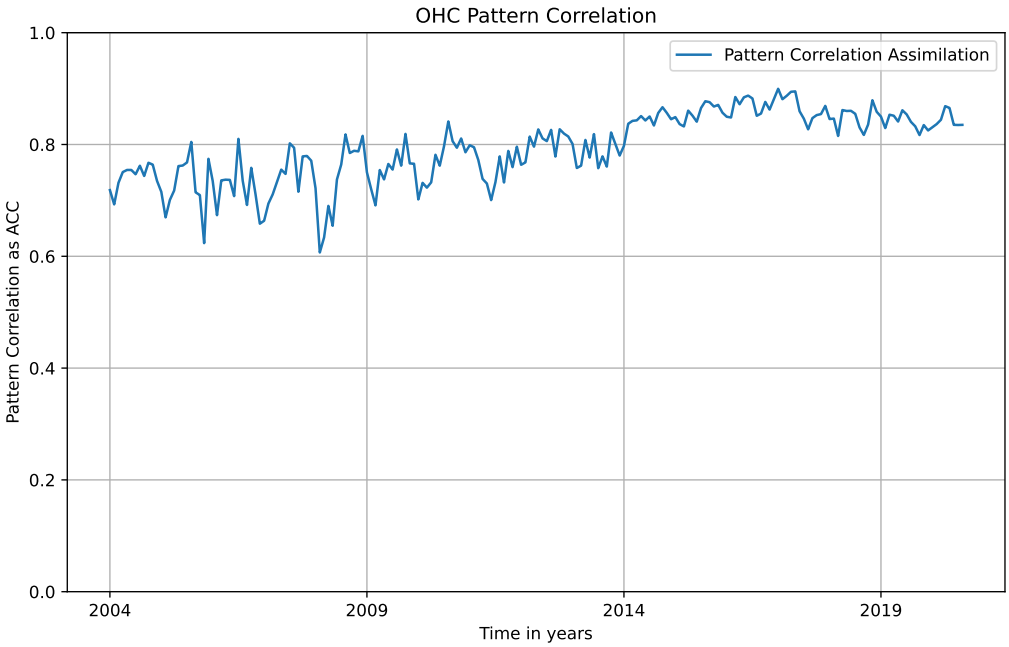

Figure 3.4: Pattern correlation for monthly estimates of anomaly NA OHC of one EnKF assimilation ensemble member and neural network reconstructions of masked assimilations.

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